IGCSE 0607 Practice Questions — E10.7 — Combined Questions & Markscheme

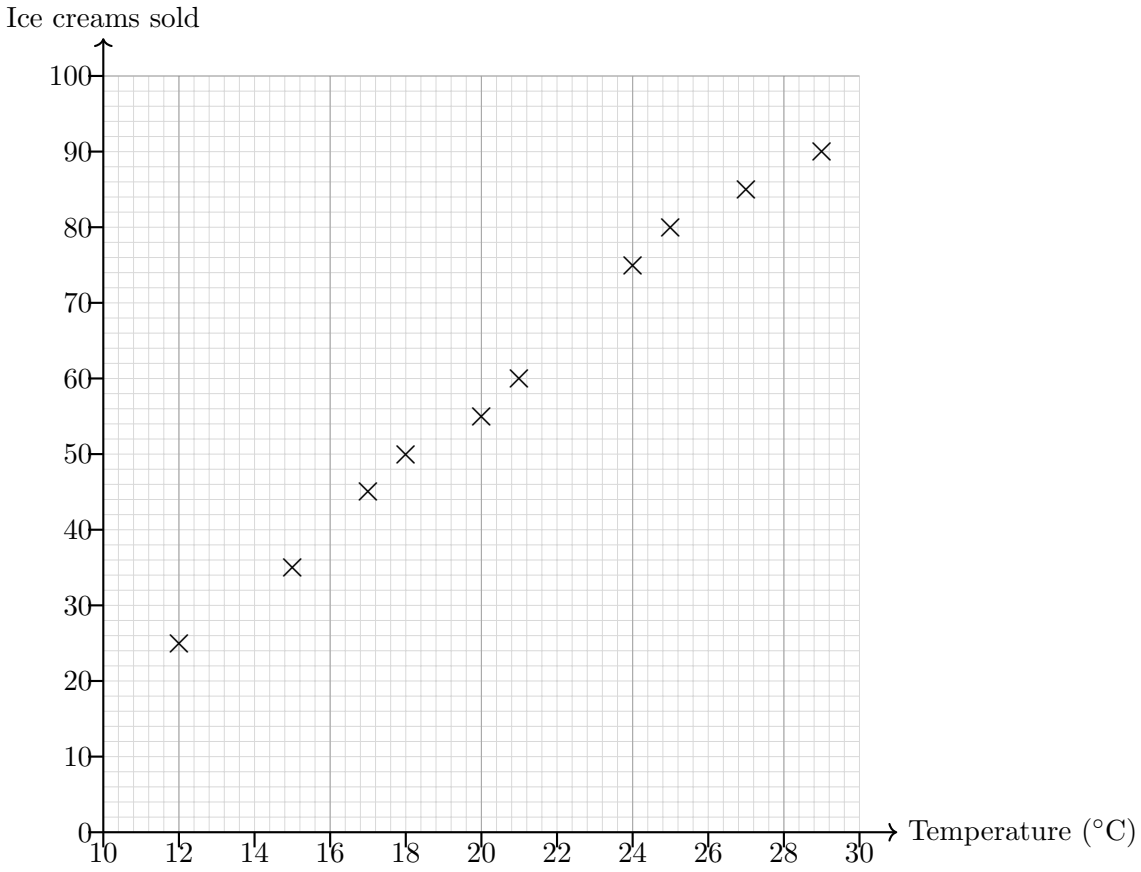
Scatter diagrams

Total Questions: 15 — **Total Marks:** 96 Marks

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- **E10.7** Scatter diagrams
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(12 Qs)*

1 The scatter diagram shows the relationship between the maximum daily temperature, x°C, and the number of ice creams sold, y, on 10 different days.



(a) State the type of correlation shown in the scatter diagram.

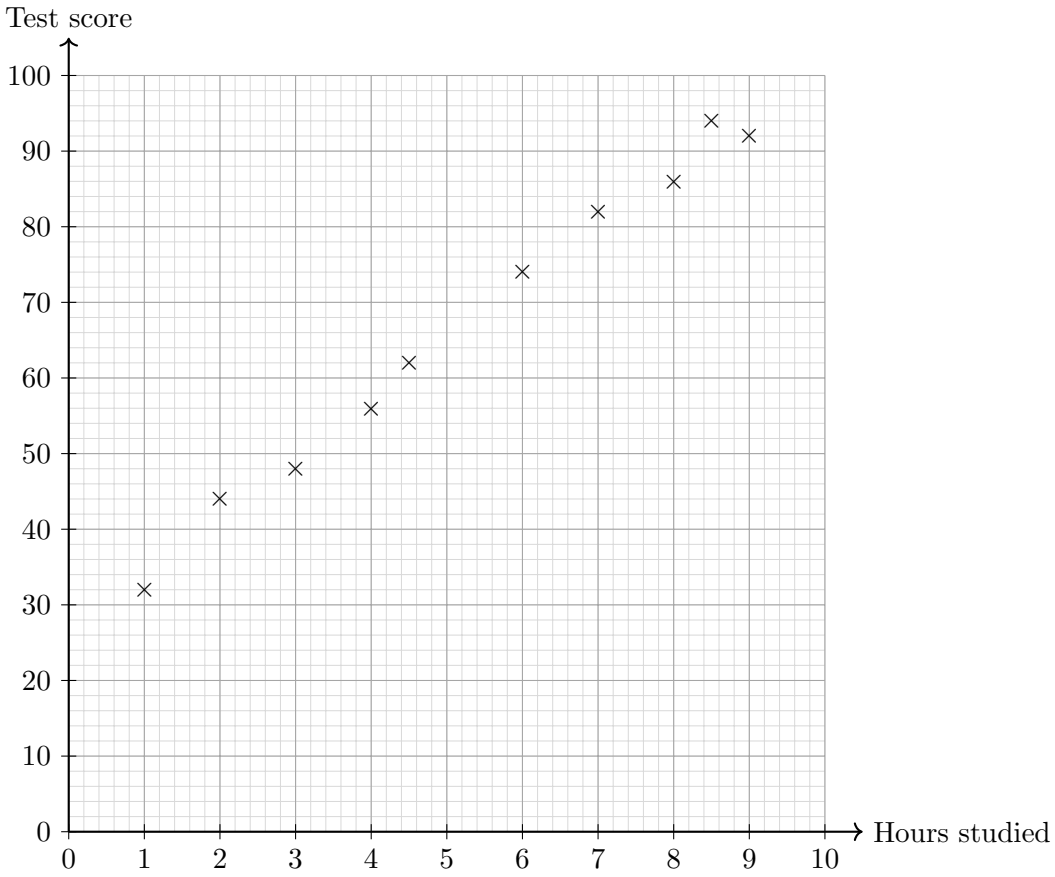
..... **[1]**

(b) On the scatter diagram, draw a line of best fit. **[1]**

(c) Use your line of best fit to estimate the number of ice creams sold on a day when the maximum temperature is 22°C.

..... **[1]**

2 The scatter diagram shows the number of hours studied and the test score achieved by 10 students.



(a) What type of correlation is shown by the scatter diagram?

..... **[1]**

(b) On the scatter diagram, draw a line of best fit. **[1]**

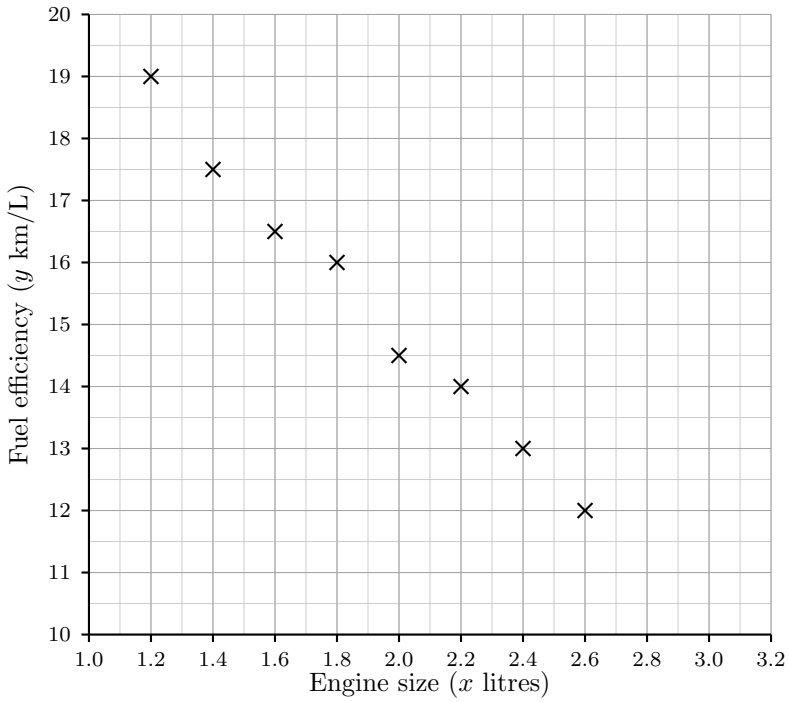
(c) A student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

3 A motoring magazine records the engine size, x litres, and the fuel efficiency, y km/L, for 10 cars. The table shows the results.

Engine size (x L)	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
Efficiency (y km/L)	19.0	17.5	16.5	16.0	14.5	14.0	13.0	12.0	11.5	11.0

(a) The first 8 points are plotted on the scatter diagram below.
Plot the remaining 2 points, $(2.8, 11.5)$ and $(3.0, 11.0)$, on the grid.



[2]

(b) State the type of correlation shown by the scatter diagram.

..... **[1]**

(c) **(i)** Calculate the mean engine size, $\bar{x}$, and the mean fuel efficiency, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

(ii) On the scatter diagram in part (a), plot the mean point $(\bar{x}, \bar{y})$ and draw a straight line of best fit by eye. **[2]**

(d) Use your line of best fit to estimate the fuel efficiency of a car with an engine size of 2.5 litres.

..... km/L **[1]**

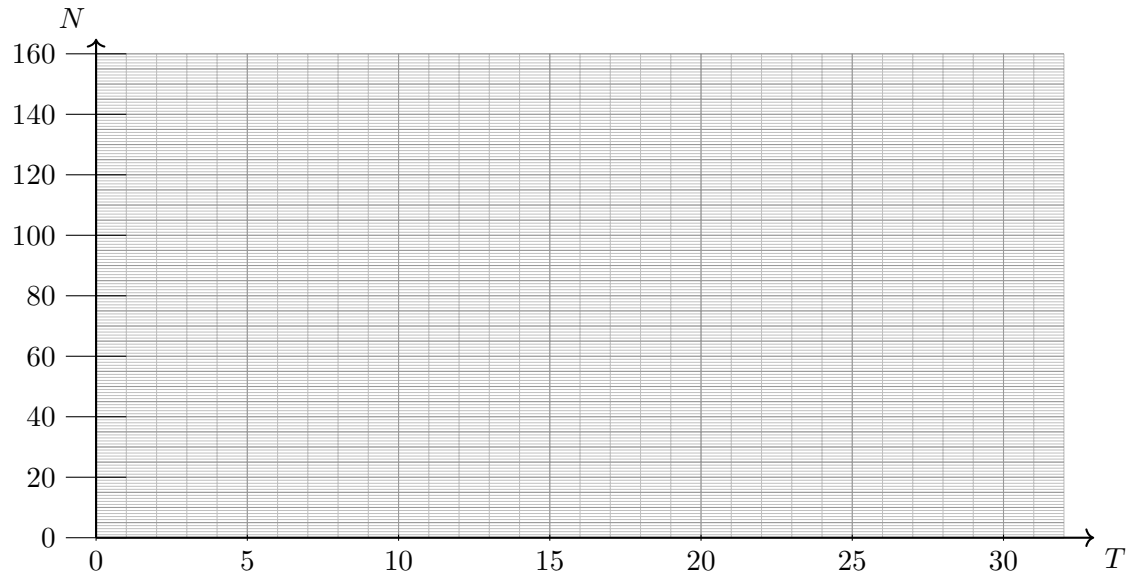
(e) Explain why your line of best fit should not be used to estimate the fuel efficiency of a car with an engine size of 5.0 litres.

..... **[1]**

4 The table shows the outside temperature, T°C, and the number of hot drinks, N, sold by a cafe on 10 different days.

Temperature (T°C)	5	8	12	15	18	20	22	25	28	30
Number of hot drinks (N)	140	125	100	95	70	65	50	40	25	15

(a) Complete the scatter diagram.
The first four points have been plotted for you.



[2]

(b) What type of correlation is shown by the scatter diagram?

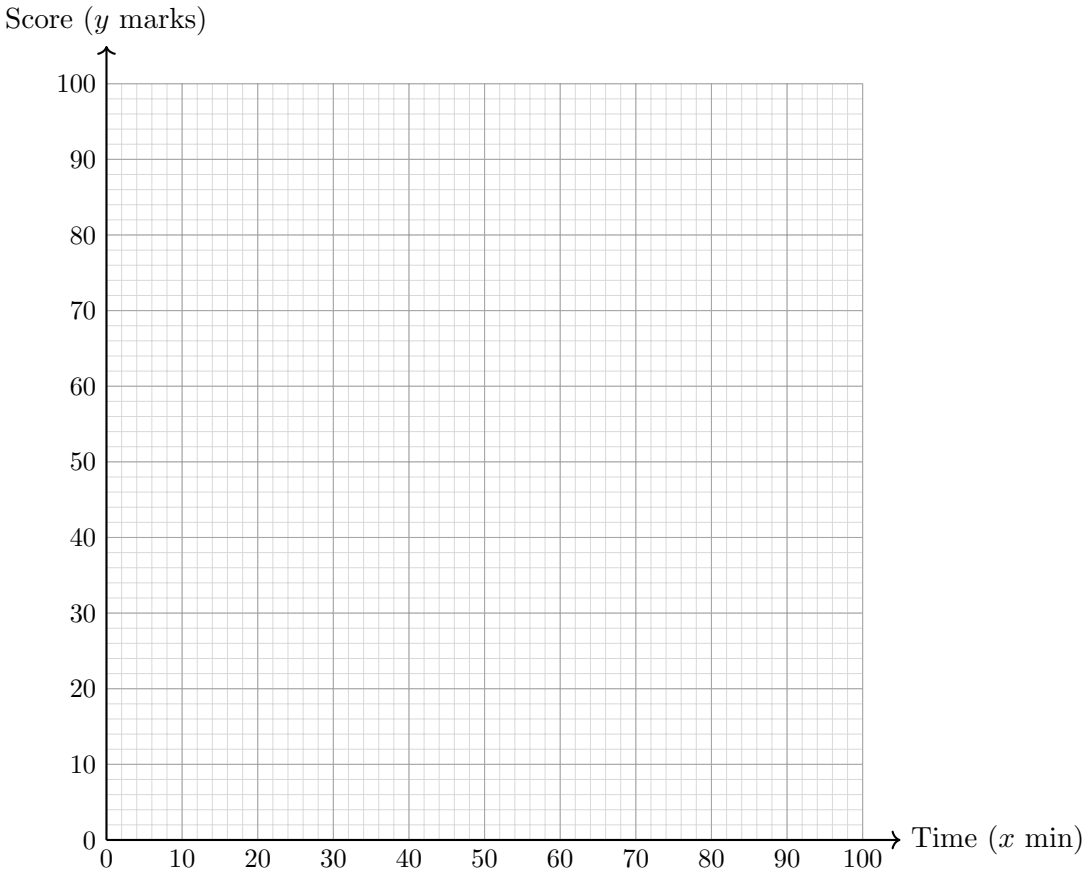
..... **[1]**

(c) Draw a line of best fit on the scatter diagram. **[1]**

5 A teacher records the time, x minutes, spent studying and the score, y marks, for 8 students.
The table shows the results.

Time (x min)	10	20	30	40	50	60	70	80
Score (y marks)	15	25	20	40	50	70	60	80

(a) On the grid, complete the scatter diagram to show this information.



[2]

(b) What type of correlation is shown by the scatter diagram?

..... **[1]**

(c) On the scatter diagram, draw a line of best fit. **[1]**

(d) Another student spent 45 minutes studying.
Use your line of best fit to estimate their score.

..... **[1]**

6 The table shows the time spent studying, x hours, and the test score, y marks, for a group of 8 students.

Study time (x hours)	2	3.5	4	5	6.5	8	9	10
Test score (y marks)	45	52	60	68	75	82	85	95

(a) Use your graphic display calculator to find the equation of the linear regression line, giving y in terms of x.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

(b) Use your equation to estimate the test score for a student who studies for 7 hours.

..... **[2]**

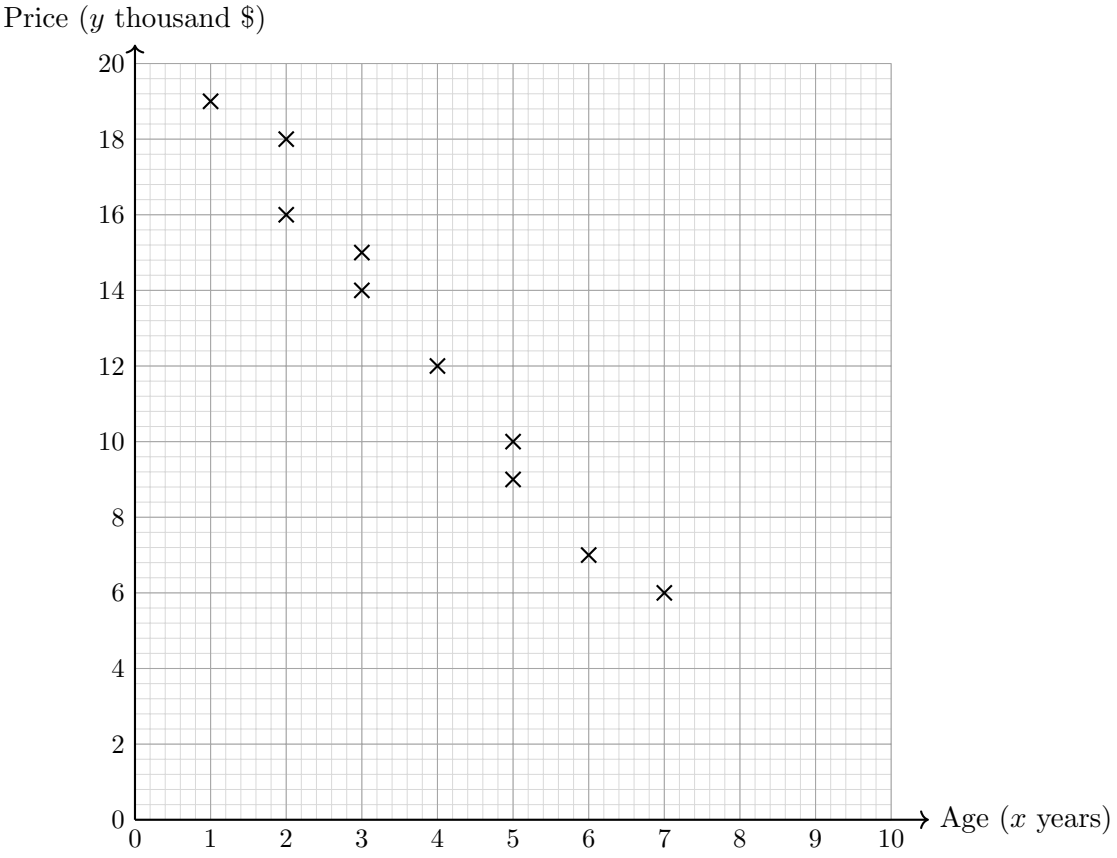
(c) Explain why it may not be reliable to use this equation to estimate the test score for a student who studies for 15 hours.

..... **[1]**

7 A car dealer records the age, x years, and the price, y thousand dollars, of 12 used cars of the same model.
The table shows the data.

Age (x years)	1	2	2	3	3	4	5	5	6	7	7	9
Price (y thousand \$)	19	18	16	15	14	12	10	9	7	6	5	1

(a) The scatter diagram shows the first 10 points from the table.
Plot the remaining 2 points on the scatter diagram.



[1]

(b) Calculate the mean age and the mean price.

Mean age = ... , Mean price = ... **[2]**

(c) On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

(d) State the type of correlation shown.

..... **[1]**

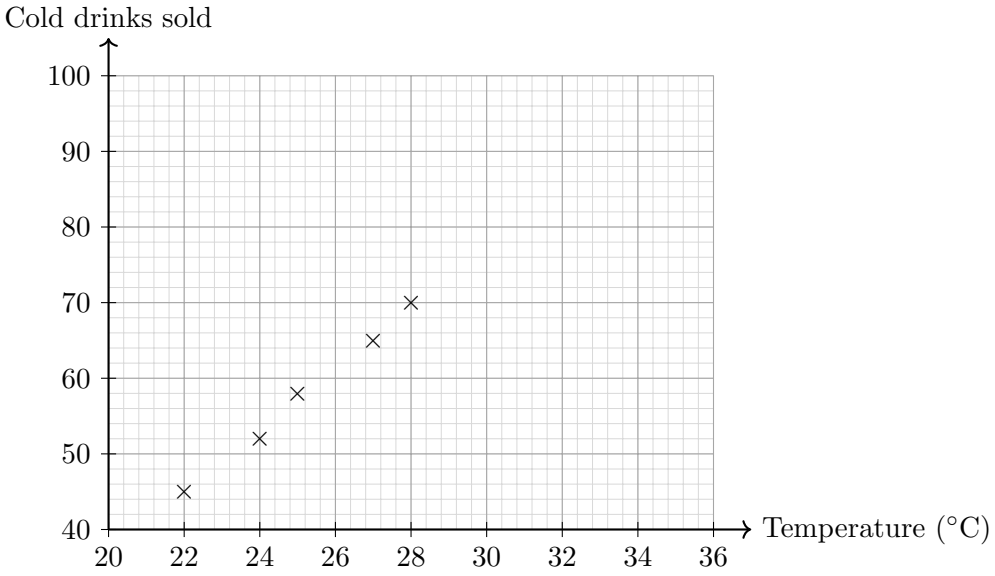
(e) Use your line of best fit to estimate the price of a car that is 8 years old.

..... thousand dollars **[1]**

8 The table shows the maximum temperature, x°C, and the number of cold drinks sold, y, at a shop on 8 different days.

Temperature, x (°C)	22	24	25	27	28	31	32	35
Cold drinks sold, y	45	52	58	65	70	81	85	96

The first 5 points have been plotted on the scatter diagram.



(a) Plot the remaining 3 points on the scatter diagram. **[2]**

(b) Calculate the mean temperature and the mean number of cold drinks sold.

Mean temperature = ... , Mean drinks sold = ... **[2]**

(c) On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

(d) Predict the number of cold drinks sold on a day when the temperature is 29°C.

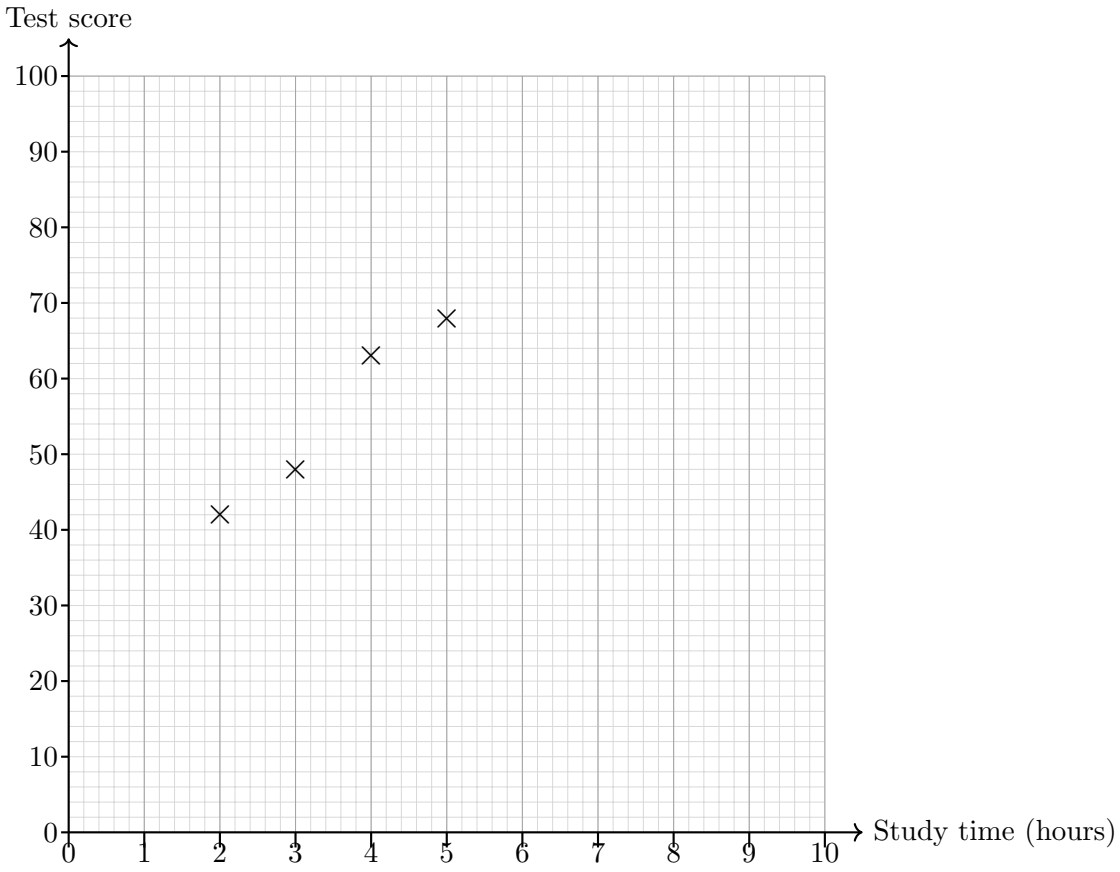
..... **[1]**

9 A teacher records the number of hours spent studying, x, and the test score, y, for 7 students.

Study time (x hours)	2	3	4	5	6	7	8
Test score (y)	42	48	63	68	81	88	100

The scatter diagram for this data is partially completed.

(a) Plot the remaining three points on the scatter diagram.



[2]

(b) Calculate the mean study time, $\bar{x}$, and the mean test score, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

(c) On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

(d) Another student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

10 A teacher records the daily revision time, x minutes, and the test score, y marks, for 8 students. The results are shown in the table below.

Revision time (x min)	10	15	20	25	30	35	40	45
Test score (y marks)	34	40	44	51	55	62	65	73

(a) (i) Use your graphic display calculator to find the equation of the linear regression line of y on x, giving the coefficients correct to 3 significant figures.

$y =$ ..... **[2]**

(ii) Write down the coordinates of the mean point $(\bar{x}, \bar{y})$.

(..... , ...) **[2]**

(b) (i) Use your regression equation to estimate the test score of a student who revises for 32 minutes per day.

..... marks **[2]**

(ii) Explain why using the regression equation to estimate the test score of a student who revises for 90 minutes per day would not be valid.

..... **[2]**

(c) A 9th student revises for 25 minutes per day and scores 51 marks.
State, with a reason, how adding this student's data affects the coordinates of the mean point $(\bar{x}, \bar{y})$.

.....

..... **[2]**

11 The table shows the age, x years, and the value, y dollars, of 6 cars of the same model.

Age, x (years)	1	2	3	4	5	6
Value, y (\$)	12600	10800	9600	8100	6300	5100

(a) Use a graphic display calculator to find the equation of the linear regression line. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

(b) Give an interpretation of the value of m in this context.

..... **[1]**

(c) Use your equation to estimate the value of a car that is 4.5 years old.

% ..... **[2]**

(d) Explain why it may not be reliable to use this equation to estimate the value of a car that is 15 years old.

..... **[1]**

12 A researcher measures the height, x cm, and the mass, y kg, of 12 adult males.
The results are summarised as follows:

$$\sum x = 2100 \qquad \sum y = 924$$

The equation of the linear regression line is $y = 0.85x - c$, where c is a constant.

(a) Calculate the mean height, $\bar{x}$, and the mean mass, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

(b) The point $(\bar{x}, \bar{y})$ lies on the linear regression line.
Find the value of c.

$c =$ ..... **[2]**

(c) A new adult male is added to the study.
His height is 180 cm.
His mass is exactly equal to the value predicted by the regression line.
Calculate the new mean mass for all 13 men.

..... kg **[2]**

13 An automotive engineer tests 8 different cars to investigate the relationship between engine size, x litres, and fuel consumption, y km/l.
The results are shown in the table.

Engine size (x litres)	1.0	1.2	1.5	1.8	2.0	2.4	2.5	3.0
Fuel consumption (y km/l)	21.5	20.0	18.2	16.5	15.0	13.0	12.5	9.8

(a) (i) Use a graphic display calculator to find the equation of the linear regression line.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

(ii) Explain what the value of the gradient represents in this context.

..... **[1]**

(b) Use your equation from part (a)(i) to estimate the fuel consumption of a car with an engine size of 1.6 litres.

..... km/l **[2]**

(c) Explain why it may not be reliable to use this equation to estimate the fuel consumption of a car with an engine size of 5.0 litres.

..... **[1]**

14 The table shows the temperature, x°C, and the number of ice creams sold, y, at a shop over 8 days.

Temperature (x°C)	12	15	18	20	22	25	28	30
Ice creams sold (y)	40	52	68	75	84	98	110	115

(a) Calculate the mean temperature and the mean number of ice creams sold.

Mean temperature = ... , Mean ice creams sold = ... **[2]**

(b) Use a graphic display calculator to find the equation of the linear regression line for this data. Give your answer in the form $y = mx + c$, with values of m and c correct to 3 significant figures.

$y =$ ..... **[2]**

(c) The shop owner predicts the number of ice creams sold on a day when the temperature is 24°C. Use your equation to estimate this number.

..... **[2]**

(d) Comment on the reliability of using your equation to predict the number of ice creams sold on a day when the temperature is 42°C.

..... **[1]**

15 A company records its monthly advertising spend, x thousand dollars, and its monthly sales, y thousand dollars, over 8 months.
Two entries in the table, p and q, are missing.
Over the 8 months, the mean advertising spend is $\bar{x} = 11$ thousand dollars and the mean monthly sales is $\bar{y} = 45$ thousand dollars.

Advertising (x thousand \$)	4	7	8	p	12	14	15	18
Sales (y thousand \$)	26	34	38	43	47	53	q	62

(a) Find the value of p and the value of q.

$p =$ ..... , $q =$ ..... **[2]**

(b) **(i)** Use your graphic display calculator to find the equation of the linear regression line of y on x in the form $y = ax + b$.

$y =$ ..... **[2]**

(ii) Interpret the value of a in the context of this question.

..... **[1]**

(c) Calculate the difference between the actual monthly sales when $x = 14$ and the monthly sales predicted by the regression line, giving your answer in dollars.

\$ ..... **[3]**

(d) **(i)** Use the regression equation to estimate the monthly advertising spend required to achieve monthly sales of \$68 000, giving your answer in thousand dollars correct to 3 significant figures.

..... thousand dollars **[2]**

(ii) Explain why this estimate may be unreliable.

..... **[1]**

1 **(a)** Positive **[B1]**

(b) Single straight ruled line passing within the bounds of the data points, with roughly equal numbers of points on either side. **[B1]**

(c) 64 (Accept values between 62 and 67 strictly based on the student's line of best fit) **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Positive **[B1]**

(b) A single ruled straight line drawn crossing the data set, passing within tolerance of the mean point $(6.3, 67)$ with roughly equal points on either side. **[B1]**

(c) Accept any value strictly read from the candidate's own line of best fit at $x = 5.5$ (expected roughly 60 to 62). **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Point $(2.8, 11.5)$ plotted accurately as a small cross **[B1]**

Point $(3.0, 11.0)$ plotted accurately as a small cross **[B1]**

(b) Negative **[B1]**

(c) **(i)** $\bar{x} = \dfrac{21.0}{10} = 2.1$ **[B1]**

$\bar{y} = \dfrac{145.0}{10} = 14.5$ **[B1]**

(ii) Mean point $(2.1, 14.5)$ plotted accurately **[B1]**

Single ruled straight line passing through $(2.1, 14.5)$ across the data range with roughly equal numbers of points on either side **[B1]**

(d) Answer in the range 12.4 to 13.0 (or follow-through from candidate's line at $x = 2.5$) **[B1]**

(e) 5.0 litres lies outside the range of the given data ($1.2 \leq x \leq 3.0$), so the linear trend may not continue (extrapolation is unreliable) **[B1]**

[Back to Question] *[Total Marks: 9]*

4 **(a)** Plotting the remaining 6 points correctly. **[P2]**

(Allow P1 for 4 or 5 points correct).

(b) Negative **[B1]**

(c) A ruled straight line of best fit drawn by eye, with a roughly equal number of points on each side. **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** All 8 points plotted correctly **[B2]**

(5, 6, or 7 points plotted correctly [B1])

(b) Positive **[B1]**

(c) Ruled line passing through the mean point $(45, 45)$ and fitting the trend of the data **[B1]**

(d) Value read correctly from their line of best fit (typically $42 - 48$) **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $y = 5.96x + 34.7$ (or $m = 5.96$, $c = 34.7$) **[B2]**

(Allow B1 for either $m = 5.96$ or $c = 34.7$ correct to 3 sf)

(b) Substitution of $x = 7$ into their equation from (a): $5.96(7) + 34.7$ **[M1]**

76.4 (or 76) **[A1ft]**

(c) The value $x = 15$ is outside the range of the given data (extrapolation). **[B1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Points $(7, 5)$ and $(9, 1)$ correctly plotted **[B1]**

(b) Mean age $= 4.5$ **[B1]**

Mean price $= 11$ **[B1]**

(c) Mean point $(4.5, 11)$ plotted accurately **[B1]**

Ruled line of best fit passing through $(4.5, 11)$ **[B1]**

(d) Negative (or strong negative) **[B1]**

(e) Value read correctly from their line of best fit (typically $2 - 4$) **[B1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** All three points correctly plotted at $(31, 81)$, $(32, 85)$, and $(35, 96)$ **[B2]**

Note: Award [B1] if exactly two points are plotted correctly.

(b) Mean temperature $= \dfrac{224}{8} = 28$ **[B1]**

Mean drinks sold $= \dfrac{552}{8} = 69$ **[B1]**

(c) Mean point $(28, 69)$ clearly plotted (e.g. marked with a cross or a circled cross). **[B1]**

A ruled straight line of best fit passing directly through their mean point, with an appropriate gradient. **[B1]**

(d) Accept any value strictly read from the candidate's own line of best fit at $x = 29$ (expected roughly 72 to 75). **[B1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** All three points $(6, 81)$, $(7, 88)$, and $(8, 100)$ plotted correctly **[B2]**

(Allow B1 if only 1 or 2 points are plotted correctly)

(b) $\bar{x} = \dfrac{35}{7} = 5$ **[B1]**

$\bar{y} = \dfrac{490}{7} = 70$ **[B1]**

(c) Mean point $(5, 70)$ correctly plotted (allow a distinct mark e.g. a circle around the cross) **[B1]**

Single straight ruled line passing exactly through their mean point, following the trend of the data **[B1]**

(d) 75 (Accept values between 73 and 77 strictly based on the student's line of best fit) **[B1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** **(i)** Gradient $a = 1.0857\ldots$ rounded to 1.09 **[B1]**

y-intercept $b = 23.1428\ldots$ rounded to 23.1, giving $1.09x + 23.1$ **[B1]**

(ii) $\bar{x} = 27.5$ **[B1]**

$\bar{y} = 53$ **[B1]**

(b) **(i)** $y = 1.0857(32) + 23.1429$ (or $1.09(32) + 23.1$) **[M1]**

57.9 (or 58) **[A1]**

(ii) 90 minutes is outside the range of the given data ($10 \leq x \leq 45$), so it involves unreliable extrapolation **[B1]**

The model gives $y \approx 121$ marks, which is impossible if the test is out of 100 (linear growth cannot continue indefinitely) **[B1]**

(c) $\bar{x}$ decreases because $25 < 27.5$ **[B1]**

$\bar{y}$ decreases because $51 < 53$ **[B1]**

[Back to Question] *[Total Marks: 10]*

11 **(a)** $m = -1500$ **[B1]**

$c = 14000$ **[B1]**

Note: Award [B2] for the fully correct equation $y = -1500x + 14000$.

(b) It represents the amount the car's value decreases (depreciates) per year. (Accept references to losing \$1500 each year). **[B1]**

(c) $y = -1500(4.5) + 14000$ **[M1]**

7250 **[A1]**

(d) 15 years is far outside the given data range (extrapolation) OR the equation would predict a negative value (-8500), which is impossible. **[B1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $\bar{x} = \dfrac{2100}{12} = 175$ **[B1]**

$\bar{y} = \dfrac{924}{12} = 77$ **[B1]**

(b) Substitution of $(\bar{x}, \bar{y})$ into the equation: $77 = 0.85(175) - c$ **[M1]**

$c = 148.75 - 77 = 71.75$ **[A1]**

(c) Predicted mass $= 0.85(180) - 71.75 = 81.25$ **[M1]**

New mean $= \dfrac{924 + 81.25}{13} = 77.3$ (or 77.32...) **[A1]**

[Back to Question] *[Total Marks: 6]*

13 (a) (i) $m = -5.82$ (accept -5.81 to -5.82) **[B1]**

$c = 27.0$ (accept 27.0 to 27.1) **[B1]**

(ii) The decrease in fuel consumption (in km/l) for every 1 litre increase in engine size. **[B1]**

(b) Substitution of $x = 1.6$ into their equation: $-5.816(1.6) + 27.0$ **[M1]**

17.7 (accept 17.6 to 17.8 depending on rounding of m and c) **[A1]**

(c) Because 5.0 is outside the range of the given data (extrapolation). **[B1]**

[Back to Question] *[Total Marks: 6]*

14 (a) Mean temperature = 21.25 **[B1]**

Mean ice creams sold = 80.25 **[B1]**

(b) $m = 4.28$ **[B1]**

$c = -10.6$ **[B1]**

(c) $y = 4.28 \times 24 - 10.6$ **[M1]**

92 (or 92.12) **[A1]**

(d) Unreliable because 42°C is outside the range of the given data (extrapolation) **[B1]**

[Back to Question] *[Total Marks: 7]*

15 (a) $p = 8(11) - (4 + 7 + 8 + 12 + 14 + 15 + 18) = 10$ **[B1]**

$q = 8(45) - (26 + 34 + 38 + 43 + 47 + 53 + 62) = 57$ **[B1]**

(b) (i) $a = 2.62$ **[B1]**

$b = 16.18$, giving $2.62x + 16.18$ **[B1]**

(ii) For every additional \$1000 spent on advertising, monthly sales increase by \$2620 on average **[B1]**

(c) Predicted sales $= 2.62(14) + 16.18 = 52.86$ thousand dollars **[M1]**

Difference $= 53 - 52.86 = 0.14$ thousand dollars **[M1]**

140 **[A1]**

(d) (i) $68 = 2.62x + 16.18$ giving $x = \dfrac{51.82}{2.62} = 19.7786\ldots$ **[M1]**

19.8 **[A1]**

(ii) Extrapolation: $x = 19.8$ lies outside the given range of advertising spend $(4 \leq x \leq 18)$ **[B1]**

[Back to Question] *[Total Marks: 11]*